

Martin Apollo Sim

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Education

University of California, Berkeley

Expected May 2029

B.S. Electrical Engineering & Computer Science

Relevant Coursework:

- Data Structures & Algorithms, Structure and Interpretation of Computer Programs, Linear Algebra & Differential Equations, Foundations of Signals, Dynamical Systems, and Information Processing, Multivariable Calculus

Experience

Lawrence Berkeley National Laboratory (ENIGMA SFA)

January 2026 - Present

Software Development Affiliate

Berkeley, CA

- Leading development of a React-based data visualization platform for microbial community analysis, replacing legacy Dash applications with a modular, production-ready frontend architecture
- Designing and integrating frontend data flows with FastAPI REST endpoints, implementing structured request/response schemas to support complex biological datasets
- Mapping relational database schemas to normalized frontend state models to maintain UI/backend data consistency
- Improving performance and user workflow efficiency by reducing redundant API calls, optimizing component re-render patterns, and restructuring data-fetch logic

Computers 4 Kids

June 2024 - August 2024

Volunteer Technician

Sacramento, CA

- Led a team of 5 interns and volunteers to restore 30+ computers, providing resources to low-income students
- Diagnosed hardware failures and performed repairs including SSD replacement, RAM upgrades, OS reinstalls
- Implemented software updates and basic system hardening techniques to ensure device security for users

Projects

Weight Lifting Tracker App

React, Node.js, Express, MongoDB

- Built and deployed a full-stack workout tracking application enabling users to log multi-set exercises and visualize per-exercise volume-load progression through interactive data visualizations
- Designed a RESTful API with Express and Mongoose supporting CRUD operations, schema validation, and password-authenticated write endpoints via middleware
- Implemented server-side aggregation pipeline to compute derived fitness metrics and return date-sorted results for real-time chart updates
- Deployed production-ready architecture with React frontend (Vercel) and Node backend (Render), configuring CORS policies, environment-based routing, and secure environment variable management for cross-origin communication

Personal Portfolio Website

TypeScript, HTML, CSS, Google Model Viewer API

- Architected a lightweight SPA using vanilla TypeScript and hash-based routing without external frameworks
- Developed reusable, modular UI components including a dynamic image carousel
- Integrated Google's model viewer web component to embed interactive, auto-rotating 3D .glb models directly in the browser with no additional tooling
- Deployed on GitHub Pages showcasing hardware and software projects

Custom Keyboard YouTube Channel - Simulator Tech

Founder/Content Creator

- Grew a YouTube Channel to 3k+ subscribers and 1M+ total views building and reviewing custom mechanical keyboards
- Secured sponsorships with 20+ companies in the mechanical keyboard industry
- Commissioned, built, and sold 15+ keyboards to clients worldwide, generating over \$1k in revenue
- Designed and Prototyped custom mechanical keyboards using Fusion360, collaborating with other keyboard designers to ensure accuracy and to bring open-source resources to other community members

Skills

- Languages & Frameworks: Python, Java, TypeScript, JavaScript, HTML, CSS, SQL, Scheme
- Frameworks & Tools: React, Node.js, Express, MongoDB
- Engineering & Hardware: Git, REST APIs, Fusion360